

MLOPS Assignment 1

Group Number 119

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MLOps Project Report: Heart Disease Prediction

1. Introduction

This report presents an end-to-end MLOps pipeline developed for predicting the presence of heart disease using clinical data. The objective of the project is not limited to building a machine learning model, but to demonstrate how such a model can be trained, tested, deployed, and monitored in a reproducible and production-ready manner.

2. Dataset and Problem Statement

The dataset used in this project is the UCI Heart Disease dataset, which contains patient-level clinical attributes such as age, sex, chest pain type, blood pressure, cholesterol levels, ECG results, and exercise-induced angina. The original dataset contains multiple target classes representing different stages of heart disease. For this project, the problem was simplified into a

binary classification task where 0 represents no heart disease and 1 represents the presence of heart disease.

3. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the distribution of features and their relationship with the target variable. Key observations included variations in heart disease occurrence across age groups, gender, and chest pain types. Correlation analysis helped identify clinically relevant features that influence the prediction outcome.

4. Feature Engineering

Numerical features were standardized using a preprocessing pipeline. All transformations were encapsulated inside a single pipeline to prevent train-serve skew and ensure consistent preprocessing during inference.

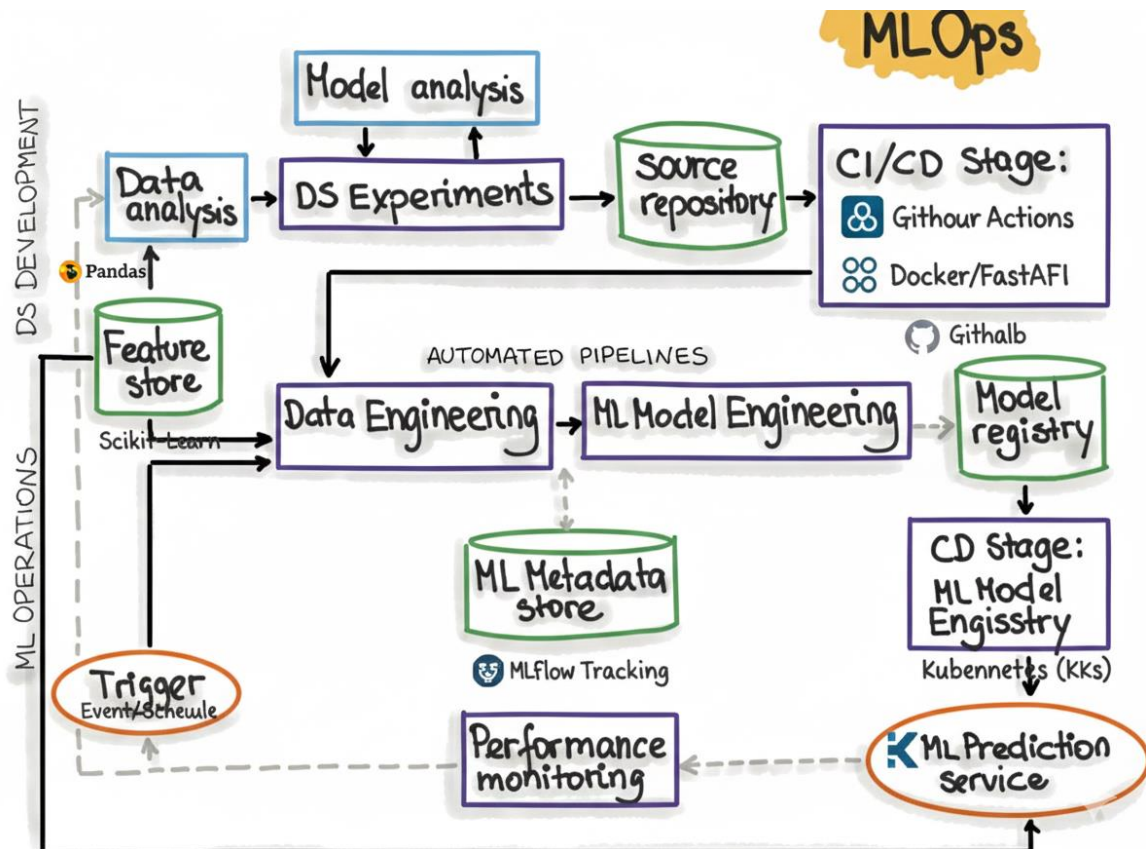
5. Model Selection and Training

Logistic Regression was selected as the primary model for this project. The choice was motivated by its interpretability, simplicity, and suitability for binary classification problems in the healthcare domain. The model was trained on preprocessed data and evaluated using standard metrics such as accuracy, precision, and recall.

MLflow was used to track experiments, including model parameters, evaluation metrics, and trained artifacts. This enabled reproducibility and easy comparison between different training runs.

6. CI/CD Pipeline and Automated Testing

A CI/CD pipeline was implemented using GitHub Actions. The pipeline is triggered on every push to the ci-cd branch. It installs dependencies, runs unit tests using Pytest, and validates the codebase. This ensures that any change to the code is automatically tested before being considered for deployment.



7. Model Serving and Containerization

The trained Logistic Regression model was served using a FastAPI-based REST API. The API exposes a /predict endpoint that accepts JSON input and returns both the prediction and a confidence score. The application was containerized using Docker, ensuring consistency across environments.

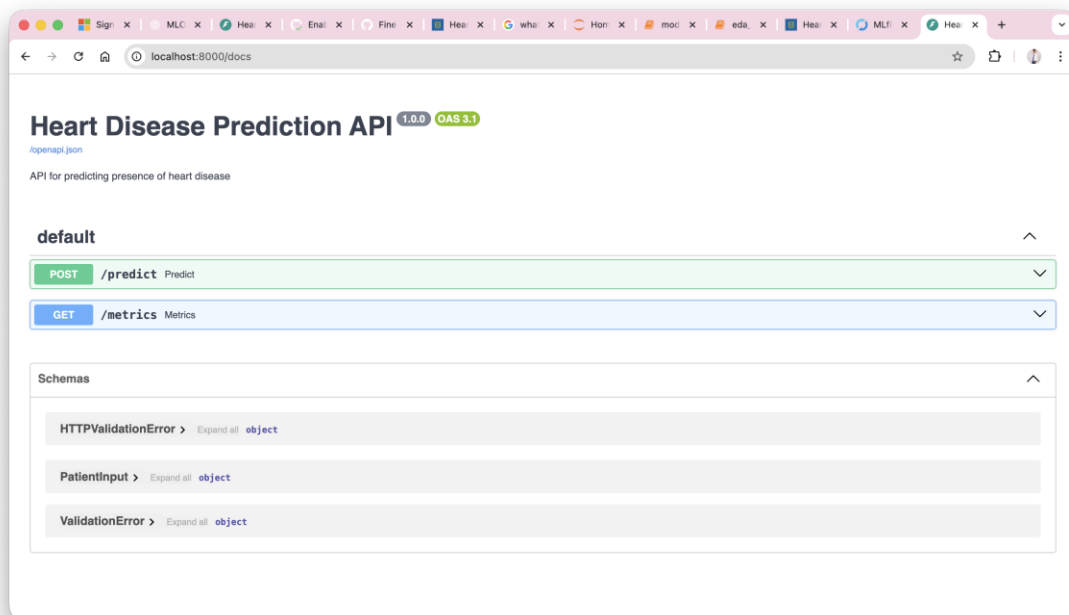
8. Deployment and Monitoring

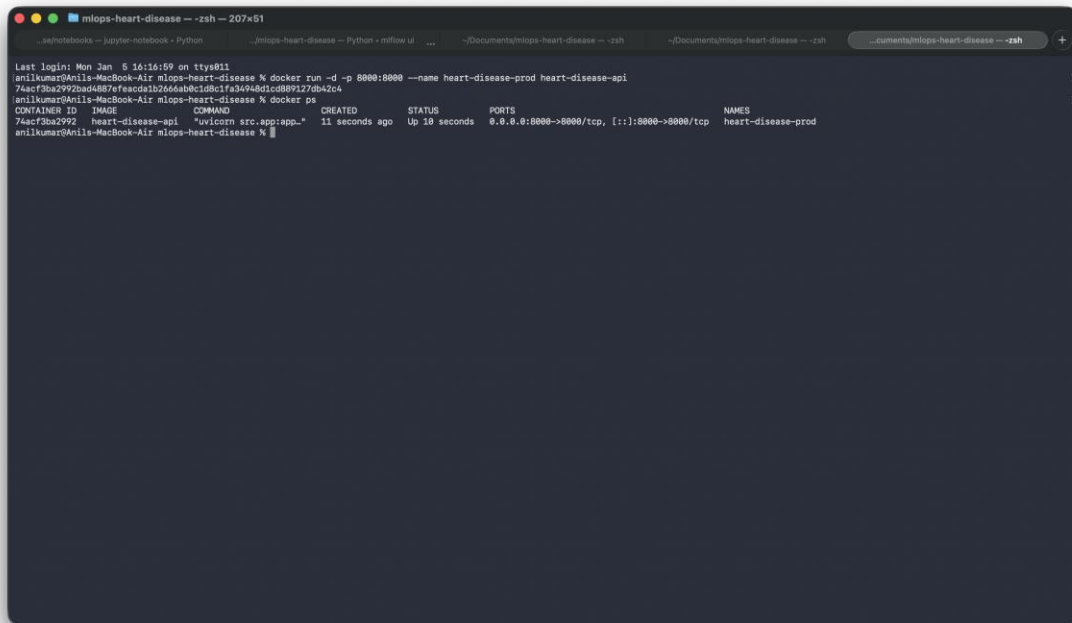
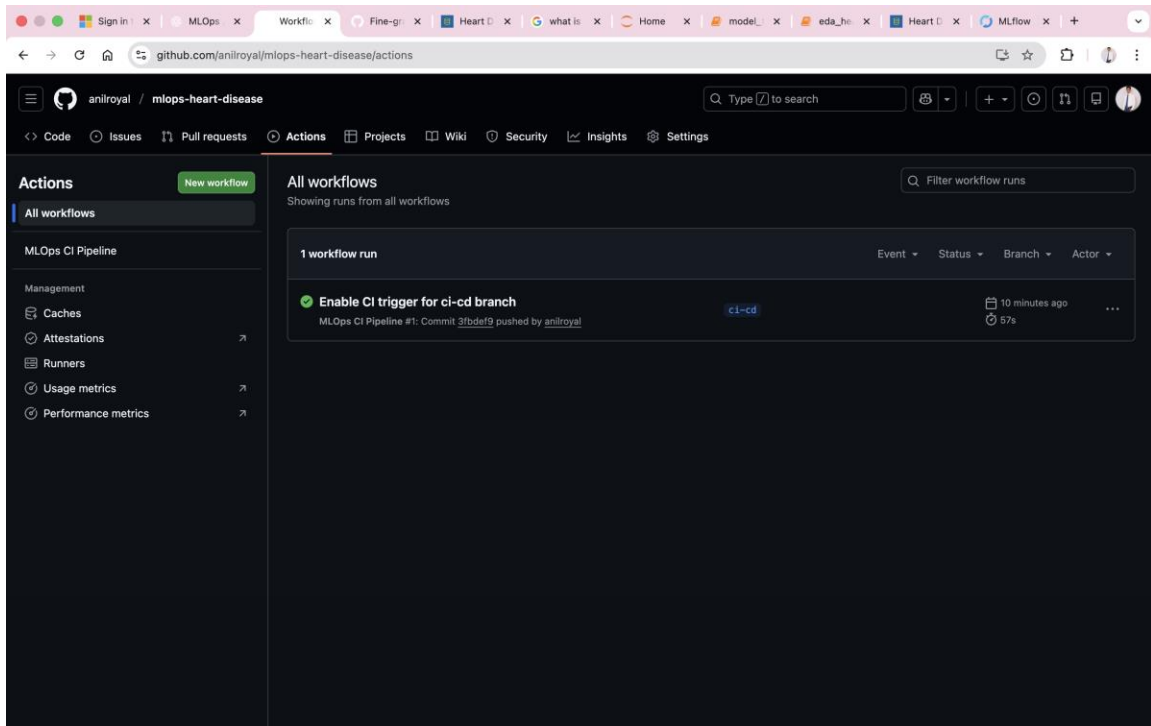
The Dockerized application was deployed locally using Docker Desktop and Kubernetes. Kubernetes deployment and service manifests were created to manage the application lifecycle and expose the API using a NodePort. Basic monitoring was implemented using Prometheus-compatible metrics such as request count and latency, exposed via a /metrics endpoint.

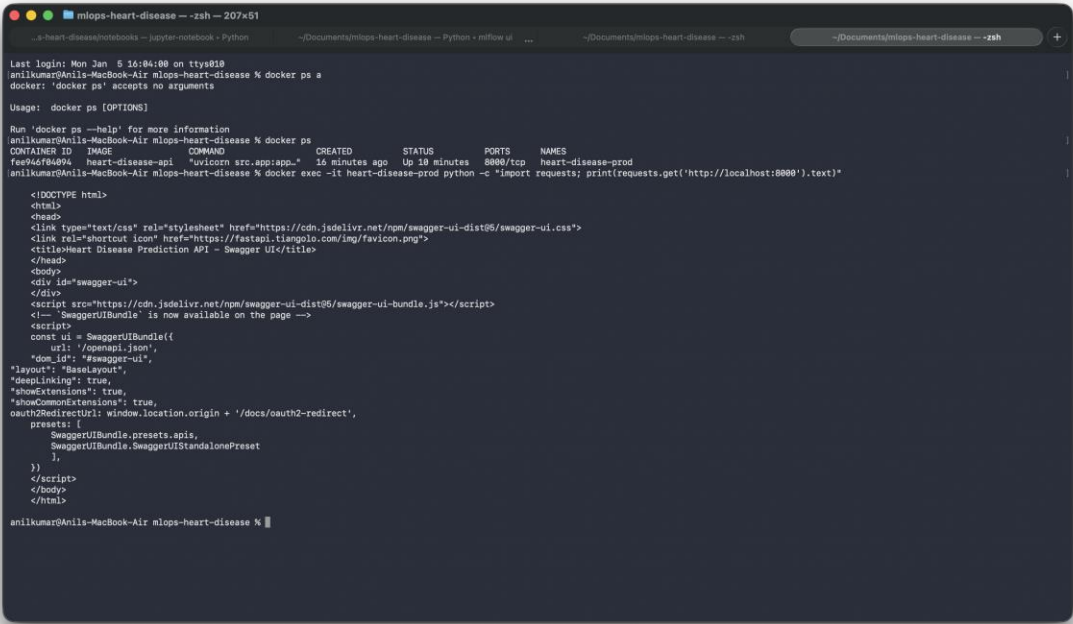
9. Conclusion

This project demonstrates a practical MLOps workflow covering data analysis, model training, CI/CD automation, containerization, deployment, and monitoring. The implementation reflects real-world practices and highlights the importance of operationalizing machine learning systems beyond model development.

Screenshots:







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ci-cd 1 Branch 0 Tags

Go to file Add file Code

Anil Kumar and Anil Kumar Enable CI trigger for ci-cd branch 3fbdef9 · 9 minutes ago 2 Commits

.github/workflows	Enable CI trigger for ci-cd branch	9 minutes ago
.idea	Initial project setup	50 minutes ago
data	Initial project setup	50 minutes ago
mlartifacts/models	Initial project setup	50 minutes ago
notebooks	Initial project setup	50 minutes ago
src	Initial project setup	50 minutes ago
test	Initial project setup	50 minutes ago
.DS_Store	Initial project setup	50 minutes ago
mlflow.db	Initial project setup	50 minutes ago
requirements.txt	Initial project setup	50 minutes ago

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← MLOps CI Pipeline

Enable CI trigger for ci-cd branch #1

Re-run all jobs

Summary

All jobs

test

Run details

Usage

Workflow file

test

succeeded 10 minutes ago in 54s

Search logs

- Set up job 8s
- Checkout code 1s
- Set up Python 1s
- Install dependencies 45s
- Run unit tests 3s
- Post Set up Python 8s
- Post Checkout code 8s
- Complete job 8s

